

Artificial Intelligence for Digital Twins: Challenges and Best Practices

Session Proposal (Workshop - 4.5h) for BNAIC/BeNeLearn 2026

Maastricht University, The Netherlands | October 21–23, 2026

Session main organisers

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Abstract:

A digital twin is a software system that tightly couples physical assets with real-time data, simulation, and AI-driven analytics, and is rapidly emerging as a foundational technology across industry, infrastructure, and science. This session aims to bring together researchers and practitioners to share recent advances in integrating AI into digital twin architectures, while critically examining the challenges, lessons learned, and best practices arising from real-world deployments. Contributors are expected to address topics such as AI-driven modeling and simulation within digital twin architectures; machine learning methods for real-time monitoring and predictive maintenance; explainable and trustworthy AI for digital twin applications; multi-agent and federated approaches for distributed digital twins; domain-specific applications in areas such as manufacturing, smart cities, healthcare, and energy; as well as evaluation frameworks, benchmarks, and emerging best practices.

Goals: Present and discuss original research on AI-enabled digital twins, identify open challenges and emerging best practices in the field, and foster community building among Benelux and international researchers.

Preliminary agenda (*leaving a buffer of ≈ 20 minutes*):

Time slot	Activity
10 min	Opening and welcome by organizers
60 min	Keynote + Q&A
80 min	Paper session 1: 4 papers (≈ 20 min)
10-15 min	Short break
80 min	Paper session 2: 4 papers (≈ 20 min)
10 min	Closure

Confirmed speakers: Joost Mertens, Post-Doctoral Researcher; University of Antwerp - Flanders Make – Tentative topic: The use of AI in the engineering of Digital Twins.

Relevance to BNAIC and AI: The application of AI to digital twins is a timely and multi-disciplinary topic that spans machine learning, knowledge representation, multi-agent systems, and human-AI interaction; all core areas of BNAIC/BeNeLearn2026. The session contributes to the conference’s aim to showcase emerging research directions and stimulate discussion on the societal impact of AI, particularly in industrial and cyber-physical contexts. This session is supported by the WEL-T ULB MLG project “Enhancing Digital Twins with Scalable and Interpretable AI (EDITS)” <https://mlg.ulb.ac.be/wel-t/>

Intended Audience: Researchers, industry practitioners, students, policymakers

Additional Considerations: Projector required

Authors of selected session papers will be invited to submit extended versions to *Digital Twins and Applications* (IET/Wiley) as the recommended publication venue, subject to the journal’s standard peer review process. Official support has been obtained from the journal.